

BTEC Level 3 Applied Science • Unit 1 Biology • Worksheet**Cell Structure**

Original JDSience practice. These questions were written for this resource and are not official exam-board items.

Learning focus: ultrastructure, membranes and structure–function relationships.

A Retrieval

1. State the function of the nucleus, a mitochondrion and a ribosome.

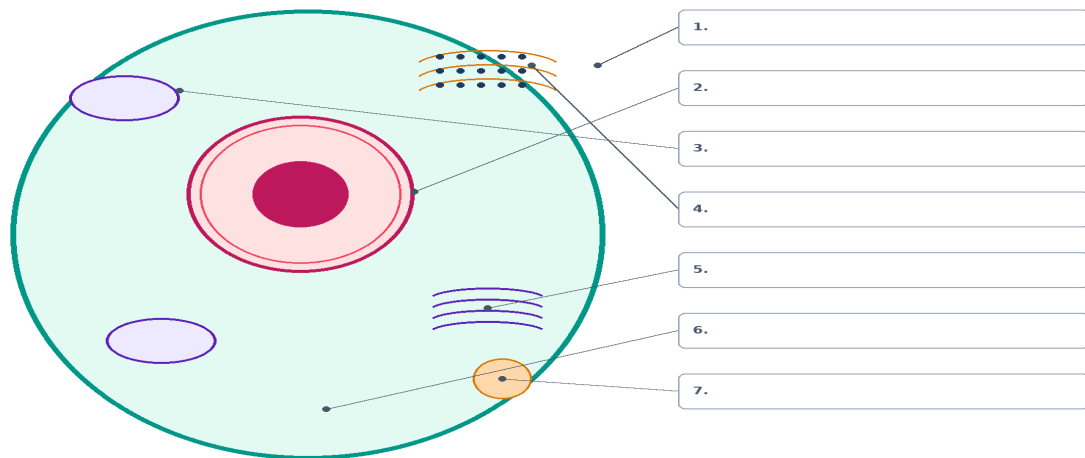
2. Name three structures found in plant cells but not in typical animal cells.

3. What is meant by a selectively permeable membrane?

B Labelling

4. On the animal-cell schematic, name the structures indicated by the numbered leader lines and state a function for each.

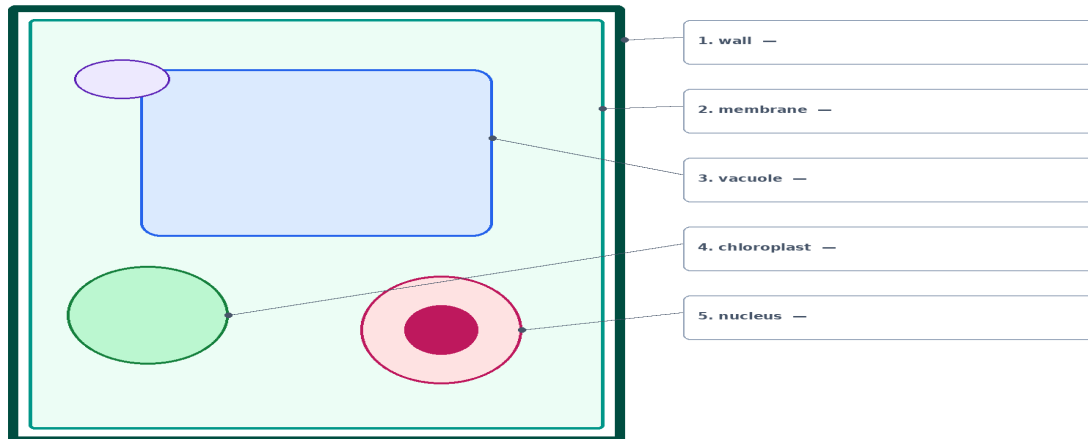
Animal cell — label the structures



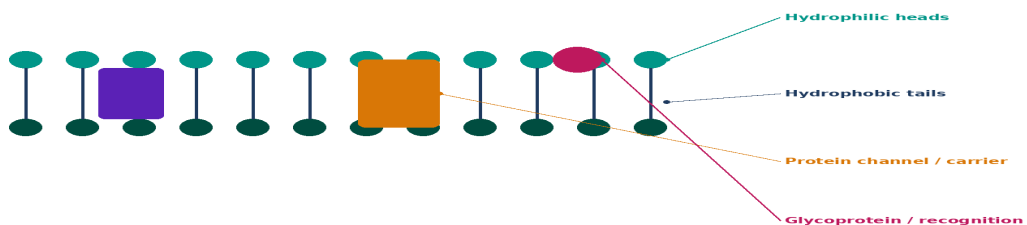
Write the name (and a function) in each box. Organelles are not to scale.

5. Label the plant-cell extras indicated by the numbered lines: wall, membrane, vacuole, chloroplast and nucleus. State which layer is freely permeable.

Plant cell — label the extra structures



Cell-surface membrane — fluid mosaic



Selectively permeable. Cholesterol (animal cells) regulates fluidity. Proteins form a mosaic.

6. Describe the structure of the cell-surface membrane. Use the words phospholipid, hydrophobic, protein and fluid mosaic.

7. Explain how cristae help mitochondria to transfer energy to ATP.

8. Why do muscle cells contain many mitochondria?

9. Root cells usually lack chloroplasts. Explain why this is not a problem for the plant.

D Exam-style practice

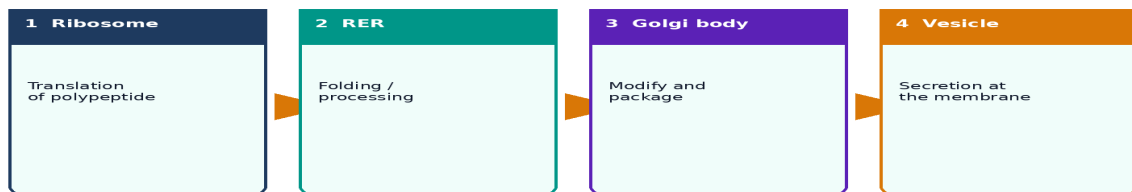
10. Describe the functions of the nucleus and mitochondria in an animal cell. [4]

11. Explain two ways a palisade cell is adapted for photosynthesis. [4]

12. Distinguish between the cell wall and the cell membrane. [3]

E Challenge

Protein trafficking: ribosome → RER → Golgi → vesicle



13. Secretory cells have extensive RER and Golgi bodies. Explain this combination in terms of protein trafficking.
